

SIMATS ENGINEERING

ASSESSMENT TOOL 2(MEDIUM)

COURSE NAME: Ethical Hacking

COURSE CODE: ITA14

COURSE OUTCOME COVERED

CO3: Analyze network services using port scanning, ping sweeps, and enumeration techniques across different operating systems to identify vulnerabilities. (BT4)

Assessment Tool Weightage: 30%

Duration: 90 min

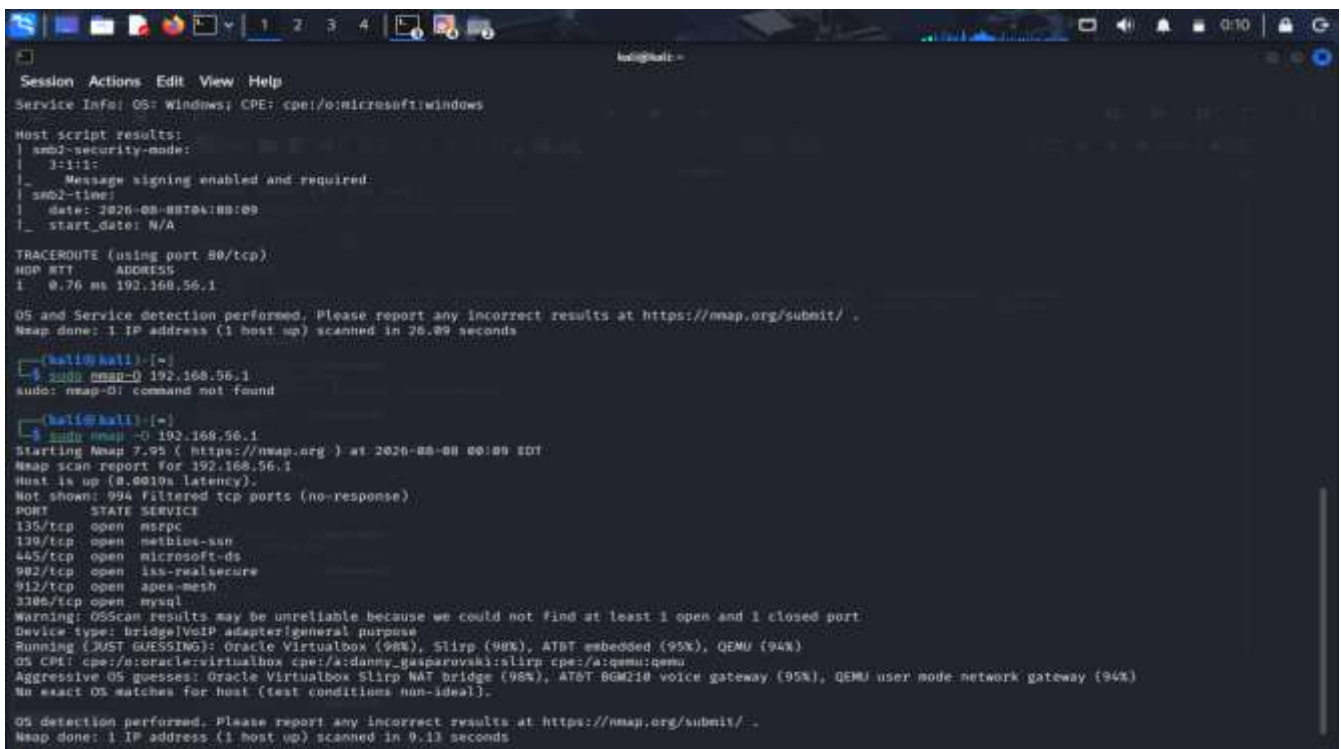
Total Marks: 50

Port Scanning Challenge

1. Using Nmap, perform a security assessment on the server 192.168.1.20. Identify all open ports and running services. Analyze the purpose of each discovered service and recommend security measures to reduce the attack surface of the system.

Security Measures

- **Port 22 (SSH):** Use strong passwords or SSH keys and restrict access.
- **Port 80 (HTTP):** Use HTTPS and disable HTTP if not required.
- **Port 443 (HTTPS):** Keep the web server and SSL/TLS configuration updated.
- **Port 3306 (MySQL):** Do not expose the database to the public network.
- **Unused ports:** Close them using a firewall.
- **All services:** Keep software updated with security patches.



```
Session Actions Edit View Help
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Host script results:
| smb2-security-mode:
|_ 3.1.1:
|_ Message signing enabled and required.
|_ smb2-time:
|_ date: 2026-08-08T04:00:00
|_ start_date: N/A

TRACEROUTE (using port 80/tcp)
HOP RTT ADDRESS
1 0.76 ms 192.168.56.1

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 26.09 seconds

kali@kali:~$ sudo nmap -i 192.168.56.1
sudo: nmap-0i command not found

kali@kali:~$ sudo nmap -i 192.168.56.1
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-08 00:00 EDT
Nmap scan report for 192.168.56.1
Host is up (0.0010s latency).
Not shown: 994 filtered tcp ports (no-response)
PORT      STATE SERVICE
135/tcp   open  msrpc
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
902/tcp   open  iis-realsecure
912/tcp   open  apex-mesh
3306/tcp   open  mysql
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Device type: bridge/vciip adapter/general purpose
Running (JUST GUESSING): Oracle Virtualbox (98%), Slirp (98%), AT&T embedded (95%), QEMU (94%)
OS CPE: cpe:/o:oracle:virtualbox cpe:/a:danny.gasparovsk:slirp cpe:/a:qemu:qemu
Aggressive OS guesses: Oracle Virtualbox Slirp NAT bridge (95%), AT&T BGM210 voice gateway (95%), QEMU user mode network gateway (94%)
No exact OS matches for host (test conditions non-ideal).

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 0.13 seconds
```

- Using Nikto, assess the web server hosted on 192.168.1.30. Identify potential web server misconfigurations, outdated components, or exposed files and directories. Discuss the security risks associated with your findings and suggest appropriate mitigation strategies.

Mitigation

- Update the web server and its components.
- Disable directory listing.
- Remove unnecessary and backup files.
- Add appropriate security headers.
- Remove default pages and configurations.
- Restrict access to sensitive directories.
- Regularly perform security assessments.

Example Findings

Finding	Risk
Outdated web server	Attackers may exploit known vulnerabilities
Directory listing enabled	Internal files and folders may be exposed
Backup files available	Sensitive source code or configuration may be revealed
Missing security headers	Provides less protection against web attacks
Default web pages	Reveals information about the server

```
kali@kali ~$ nmap -sT -oN connect_scan.txt 192.168.56.1
Nmap scan report for 192.168.56.1
Host is up (0.0031s latency).
Not shown: 994 filtered tcp ports (no-response)
PORT      STATE SERVICE
135/tcp    open  msrpc
139/tcp    open  netbios-ssn
445/tcp    open  microsoft-ds
902/tcp    open  iss-realsecure
912/tcp    open  apex-mesh
3306/tcp    open  mysql

Nmap done: 1 IP address (1 host up) scanned in 4.43 seconds

kali@kali ~$ nmap -sT -oN syn_scan.txt 192.168.56.1
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-08 00:29 EDT
Nmap scan report for 192.168.56.1
Host is up (0.0047s latency).
Not shown: 994 filtered tcp ports (no-response)
PORT      STATE SERVICE
135/tcp    open  msrpc
139/tcp    open  netbios-ssn
445/tcp    open  microsoft-ds
902/tcp    open  iss-realsecure
912/tcp    open  apex-mesh
3306/tcp    open  mysql

Nmap done: 1 IP address (1 host up) scanned in 4.95 seconds

kali@kali ~$ diff syn_scan.txt connect_scan.txt
1c1
< Nmap 7.95 scan initiated Sat Aug  8 00:29:19 2026 as: /usr/lib/nmap/nmap -sT -oN syn_scan.txt 192.168.56.1
> Nmap 7.95 scan initiated Sat Aug  8 00:29:18 2026 as: /usr/lib/nmap/nmap --privileged -sT -oN connect_scan.txt 192.168.56.1
1c1
< Host is up (0.0031s latency).
> Host is up (0.0047s latency).
13c13
< Nmap done at Sat Aug  8 00:29:23 2026 -- 1 IP address (1 host up) scanned in 4.43 seconds
> Nmap done at Sat Aug  8 00:29:23 2026 -- 1 IP address (1 host up) scanned in 4.95 seconds
```

3. Using Netcat (nc), investigate the host 192.168.1.40 by connecting to open ports and collecting available service banners. Analyze the information obtained and explain how attackers could use banner information during the reconnaissance phase of an attack.

Target Host: 192.168.1.40

Objective

To use Netcat (nc) to connect to open ports on the host and collect service banner information.

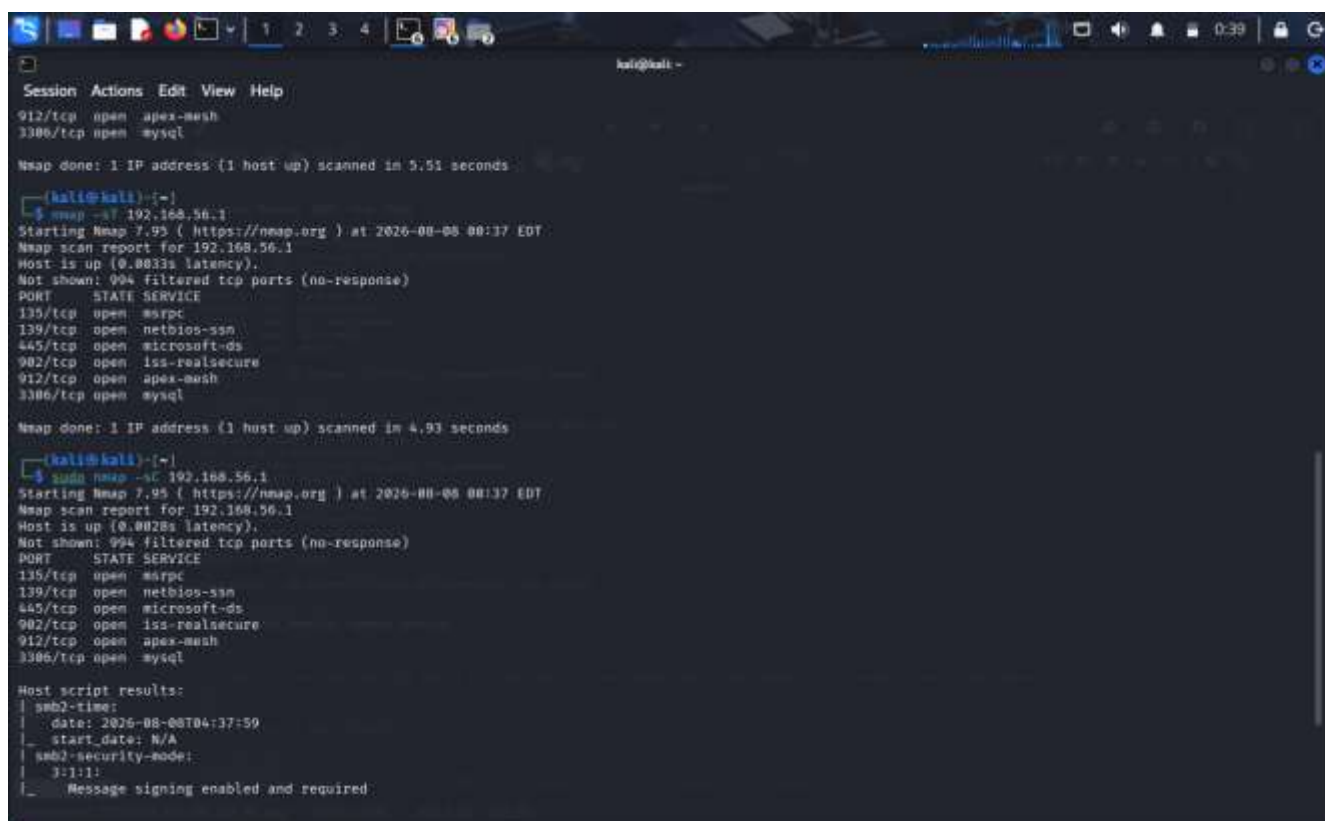
Netcat Command

```
nc -v 192.168.1.40 22
```

The same command can be used with other open ports:

```
nc -v 192.168.1.40 80
```

```
nc -v 192.168.1.40 21
```



```
kali@kali: ~  
Session Actions Edit View Help  
912/tcp open apex-mesh  
3306/tcp open mysql  
Nmap done: 1 IP address (1 host up) scanned in 5.51 seconds  
(kali@kali)-[~]  
$ nmap -sT 192.168.56.1  
Starting Nmap 7.93 ( https://nmap.org ) at 2026-08-08 00:37 EDT  
Nmap scan report for 192.168.56.1  
Host is up (0.0033s latency).  
Not shown: 994 filtered tcp ports (no-response)  
PORT      STATE SERVICE  
135/tcp   open  msrpc  
139/tcp   open  netbios-ssn  
445/tcp   open  microsoft-ds  
902/tcp   open  iss-realsecure  
912/tcp   open  apex-mesh  
3306/tcp   open  mysql  
Nmap done: 1 IP address (1 host up) scanned in 4.93 seconds  
(kali@kali)-[~]  
$ sudo nmap -sC 192.168.56.1  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-08 00:37 EDT  
Nmap scan report for 192.168.56.1  
Host is up (0.0028s latency).  
Not shown: 994 filtered tcp ports (no-response)  
PORT      STATE SERVICE  
135/tcp   open  msrpc  
139/tcp   open  netbios-ssn  
445/tcp   open  microsoft-ds  
902/tcp   open  iss-realsecure  
912/tcp   open  apex-mesh  
3306/tcp   open  mysql  
Host script results:  
_ smb2-time:  
  |_ date: 2026-08-08T04:37:59  
  |_ start_date: N/A  
_ smb2-security-mode:  
  |_ 3.1.1:  
    Message signing enabled and required
```

4. Using Telnet, examine the server 192.168.1.50 to determine whether remote administration services are accessible. Identify any security concerns related to the discovered services and recommend more secure alternatives where appropriate.

Target Server: 192.168.1.50

Objective

To use Telnet to check whether remote administration services are accessible on the server and identify related security concerns.

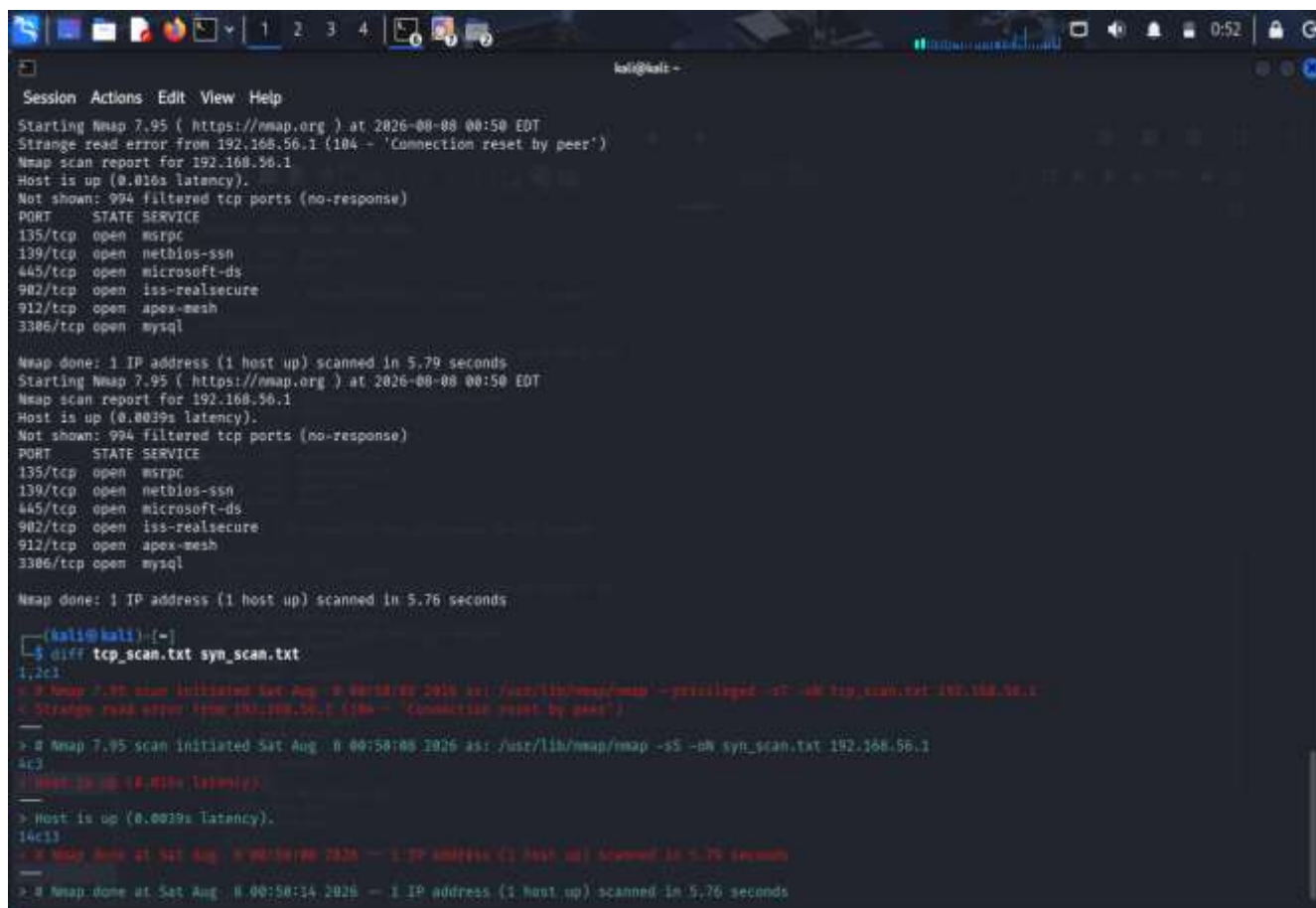
Telnet Command

```
telnet 192.168.1.50 23
```

If the connection is successful, it indicates that the **Telnet service is accessible**.

Security Concerns

Telnet is an insecure protocol because it sends login credentials and other information in **plain text**. This means an attacker who can monitor the network may be able to capture sensitive information.



```
kali@kali:~$ nmap -sT 192.168.56.1
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-08 00:50 EDT
Strange read error from 192.168.56.1 (184 - 'Connection reset by peer')
Nmap scan report for 192.168.56.1
Host is up (0.010s latency).
Not shown: 994 filtered tcp ports (no-response)
PORT      STATE SERVICE
135/tcp    open  msrpc
139/tcp    open  netbios-ssn
445/tcp    open  microsoft-ds
982/tcp    open  iss-realsecure
912/tcp    open  apex-mesh
3386/tcp   open  mysql

Nmap done: 1 IP address (1 host up) scanned in 5.79 seconds
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-08 00:50 EDT
Nmap scan report for 192.168.56.1
Host is up (0.0039s latency).
Not shown: 994 filtered tcp ports (no-response)
PORT      STATE SERVICE
135/tcp    open  msrpc
139/tcp    open  netbios-ssn
445/tcp    open  microsoft-ds
982/tcp    open  iss-realsecure
912/tcp    open  apex-mesh
3386/tcp   open  mysql

Nmap done: 1 IP address (1 host up) scanned in 5.76 seconds

kali@kali:~$ diff tcp_scan.txt syn_scan.txt
1,2c1
< # Nmap 7.95 scan initiated Sat Aug  8 00:50:00 2026 as: /usr/bin/nmap --privileged -sT -oN tcp_scan.txt 192.168.56.1
< Strange read error from 192.168.56.1 (184 - 'Connection reset by peer')
--
> # Nmap 7.95 scan initiated Sat Aug  8 00:50:00 2026 as: /usr/lib/nmap/nmap -sS -oN syn_scan.txt 192.168.56.1
4c3
< Host is up (0.010s latency).
--
> Host is up (0.0039s latency).
14c13
< # Nmap done at Sat Aug  8 00:50:00 2026 -- 1 IP address (1 host up) scanned in 5.79 seconds
--
> # Nmap done at Sat Aug  8 00:50:14 2026 -- 1 IP address (1 host up) scanned in 5.76 seconds
```

5. Using Netdiscover, perform host discovery on the network segment 192.168.1.1–50. Identify active hosts on the network and analyze how the discovered information could assist an attacker in planning further enumeration or exploitation activities..2. Vulnerability Correlation (Discovery)

Objective

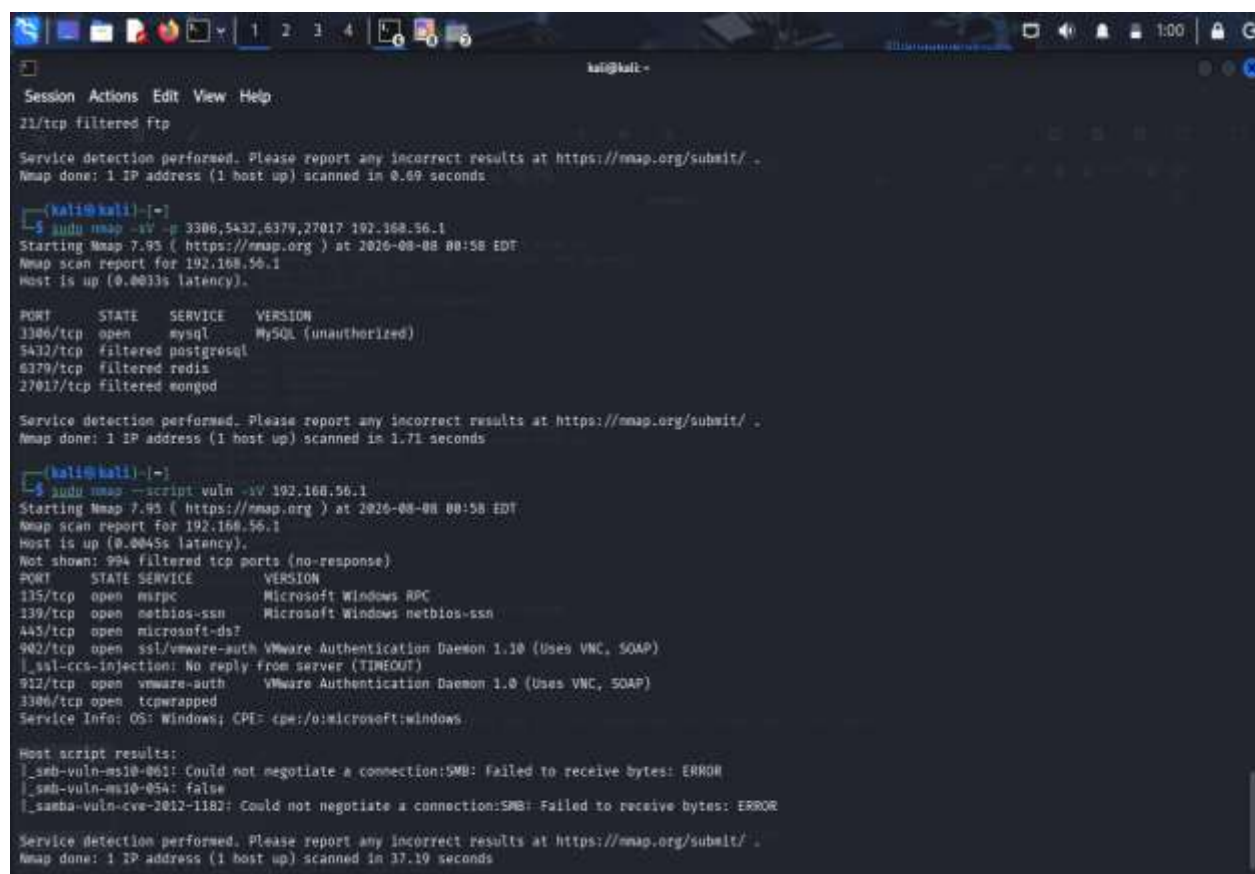
To use Netdiscover to identify active hosts in the network range 192.168.1.1–50 and understand how the discovered information can be used for further security assessment.

Netdiscover Command

```
netdiscover -r 192.168.1.0/26
```

Netdiscover identifies active devices by displaying information such as:

- IP address
- MAC address
- MAC vendor
- Active hosts on the network



```
kali@kali:~$ sudo nmap -sV -p 3306,5432,6379,27017 192.168.56.1
Starting Nmap 7.93 ( https://nmap.org ) at 2026-08-08 00:58 EDT
Nmap scan report for 192.168.56.1
Host is up (0.0033s latency).

PORT      STATE SERVICE      VERSION
3306/tcp   open  mysql        MySQL (unauthorized)
5432/tcp   filtered postgresql
6379/tcp   filtered redis
27017/tcp  filtered mongod

Service detection performed. Please report any incorrect results at https://nmap.org/submit/.
Nmap done: 1 IP address (1 host up) scanned in 0.69 seconds

kali@kali:~$ sudo nmap --script vuln -sV 192.168.56.1
Starting Nmap 7.93 ( https://nmap.org ) at 2026-08-08 00:58 EDT
Nmap scan report for 192.168.56.1
Host is up (0.0045s latency).
Not shown: 994 filtered tcp ports (no-response)
PORT      STATE SERVICE      VERSION
135/tcp   open  msrpc        Microsoft Windows RPC
139/tcp   open  netbios-ssn  Microsoft Windows netbios-ssn
445/tcp   open  microsoft-ds?
902/tcp   open  ssl/vmware-auth VMware Authentication Daemon 1.10 (Uses VNC, SOAP)
|_ssl-ccs-injection: No reply from server (TIMEOUT)
912/tcp   open  vmware-auth  VMware Authentication Daemon 1.0 (Uses VNC, SOAP)
3306/tcp   open  tcpwrapped
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Host script results:
|_smb-vuln-ms10-061: Could not negotiate a connection:SMB: Failed to receive bytes: ERROR
|_smb-vuln-ms10-054: false
|_samba-vuln-cve-2012-1182: Could not negotiate a connection:SMB: Failed to receive bytes: ERROR

Service detection performed. Please report any incorrect results at https://nmap.org/submit/.
Nmap done: 1 IP address (1 host up) scanned in 37.19 seconds
```

Code of Conduct:

I T. Nikhil Kumar Reddy (192372024) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

T. Nikhil Kumar Reddy

Q.No	Identification of Port Scanning Factors(25)	Analysis(25)	Linkage(20)	Application(20)	Clarity(10)	Marks
1						
2						
3						
4						
5						

Signature of Faculty

Total Avg :

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Identification of Port Scanning Factors	25%	Clearly identifies comprehensive and relevant strengths, weaknesses, opportunities, and threats	Identifies most factors with minor gaps	Limited or partially relevant identification	Incorrect or incomplete Port Scanning factors
Analysis	25%	Provides deep, insightful analysis of each Port Scanning component with strong reasoning	Provides good analysis with some depth	Basic analysis with limited reasoning	No meaningful analysis provided
Linkage	20%	Effectively links Port Scanning elements to derive strong strategic insights and implications	Shows some linkage with moderate insights	Weak linkage; limited strategic thinking	No linkage or strategic insight

Application	20%	Applies Port Scanning analysis effectively to the given case or scenario with clear relevance	Applies analysis with minor gaps	Limited application; lacks clarity	No clear application to context
Clarity	10%	Well-structured, clear, and logically presented with proper formatting	Generally clear with minor issues	Some structure but lacks clarity	Poorly organized and difficult to understand